

Research Methods

Day 10 · Synthesizing Findings & Polishing the Middle

Outline for Today

1. **Methods + Results Presentations: "The Middle"**
2. **Results vs. Discussion vs. Conclusion**
3. **Synthesizing Results with the Literature**
4. **Synthesizing Findings Mission Sheet**
5. **Scientific Storytelling: Discussion Slides Draft**

Welcome to Day 10!

Today you **synthesize** — turning a pile of results into a coherent story, and connecting it back to the literature.

By the end of the day you will be able to:

- **Synthesize** your results with existing literature into a coherent interpretation
- **Differentiate** results from discussion and conclusion
- **Strengthen** your methods and results sections
- Understand the role of **limitations** in scientific humility



"**The Middle**" of your paper is where data becomes meaning.



A finding only matters once you say **what it means** and **how it fits** what others have found.



Methods + Results Presentations: "The Middle"

Methods + Results Presentations 40 min

Each team presents its **Methods** and **Results** slides — same structured format: **2 min present + 2 min feedback.**

As the audience, focus on three questions:

- 1 Are the **results** clear?
- 2 Are the **figures** interpretable?
- 3 Is the **methods** description accurate?

 Feedback names **one strength** and **one specific fix** — "the axis needs units," not "nice chart."

40:00



Results vs. Discussion vs. Conclusion

Three Different Jobs

The biggest write-up confusion: blurring three sections that each do **one** job.

Results

What you found. Facts and numbers, no opinion. *"Group A scored 8 points higher ($p = 0.01$)."*

Discussion

What it means. Interpret, compare to the literature, explain, note limits. *"This aligns with Lee (2020)..."*

Conclusion

So what? The big-picture takeaway and why it matters. Short, no new data.

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Quick test: a sentence with a **number** is usually a *result*, a sentence with "**this suggests** / **because** / **compared to**" is usually *discussion*.

Spot the Section

Which section does each sentence belong in — **Results**, **Discussion**, or **Conclusion**?

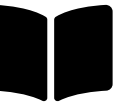
1. *"Students who slept more scored 6 points higher on average."*
2. *"This supports Garcia (2019), who linked sleep to memory consolidation."*
3. *"Schools should consider later start times to support learning."*

Spot the Section

Which section does each sentence belong in — **Results**, **Discussion**, or **Conclusion**?

1. *"Students who slept more scored 6 points higher on average."*
2. *"This supports Garcia (2019), who linked sleep to memory consolidation."*
3. *"Schools should consider later start times to support learning."*

1.  **Results** — a measured fact.
2.  **Discussion** — compares to the literature.
3.  **Conclusion** — the broader "so what."



Synthesizing Results with the Literature

What Does "Discuss" Actually Mean? 25 min

To **discuss** your findings is to do **five things, in order**:

- 1 **Restate** your key finding (in plain language).
- 2 **Compare** it to what the literature said — agree? disagree? extend?
- 3 **Explain** *why* you might have found what you found.
- 4 **Acknowledge** your limitations.
- 5 **Suggest** what comes next.

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This is the skeleton of **every** Discussion section you'll ever read. Master the order and the writing gets much easier.

Synthesis in Action

A worked Discussion mini-paragraph:

*"We found that students with more screen time slept about 1.3 hours less (**restate**). This echoes Twenge (2017), who reported the same link in a national sample (**compare**). A likely mechanism is delayed bedtime from late-night use (**explain**). Our sample was small and self-reported (**limitation**), so a larger study with objective sleep tracking is a natural next step (**next**)."*

Every finding earns a citation.

Each result in your Discussion should connect to **at least one source** from your literature review — agreement *or* tension both count.

Limitations: Scientific Humility

Naming limitations isn't weakness — it's **credibility**. It shows you understand what your study can and cannot claim.

Common, honest limitations:

- Small or non-random **sample**
- **Self-reported** or noisy measures
- **Observational** data → correlation, not proof of cause
- Short **time window**

Frame each limitation forward:

"Because our sample was one school, results may not generalize — a multi-school study would test this."

 A limitation **bounds** your claim; it doesn't erase it.



Synthesizing Findings Mission Sheet

Session 10 Mission Sheet 35 min

As a team, complete the "**Synthesizing Findings**" Mission Sheet — four sentences.

- 1 **Main conclusion** — your single biggest takeaway.
- 2 **Comparison to literature** — one source you agree or disagree with.
- 3 **One important limitation.**
- 4 **One next step.**

Session 10 Mission Sheet 35 min

As a team, complete the "**Synthesizing Findings**" Mission Sheet — four sentences.

- 1 **Main conclusion** — your single biggest takeaway.
- 2 **Comparison to literature** — one source you agree or disagree with.
- 3 **One important limitation.**
- 4 **One next step.**

 **Swap with a neighboring team** for feedback before you finalize.

35:00



Scientific Storytelling: Discussion Slides Draft

Draft Your Discussion & Conclusion Slides 20 min

Using the Session 7 "**Presenting Research: The Ending**" Mission Sheet as your guide, draft your **Discussion** and **Conclusion** slides.

Answer three key questions:

- 1 What do our results **mean**?
- 2 Did they **support or challenge** our hypothesis?
- 3 Who should **care** about this?

✓ A strong Discussion slide **adds interpretation** — it never just repeats what the chart already shows.

20:00



Homework & After-Class Work

After-Class Team Session ⚙️ 3:15 – 5:15 PM

Teams work with a **TA / peer mentor** available. Produce a **tangible output before dinner** – *not free time*.

1

Session 10 Mission Sheet: write your synthesis – conclusion sentence, comparison to one literature source, one limitation, one next step.

2

Draft the Discussion slide: "What do our results mean?" – connect each finding to at least one source from your literature review.

3

DataCORE: run any remaining analyses; export all final charts as PNG with descriptive names (e.g. `fig1_income_health_scatter.png`).

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🚩 **Team output due before dinner:** draft Discussion slide + completed Session 10 Mission Sheet.

Evening Work Period 📅 7:00 – 9:00 PM

Individual + partner work during CTD RTA study hours. TA available for drop-in questions.

- **Individual:** draft your **Discussion paragraph** — did results support your hypothesis? How do findings compare to the literature? (**200–250 words**)
- **(RcS):** add a **Discussion/Conclusion** section to your written report with **≥ 2 literature comparisons (with citations)**. Target **300–400 words**.

Team slide check (*async is fine*):

- Are all charts **labeled**?
- Does the **story flow**?
- Identify **3 specific things** to fix in tomorrow's session.